

TailorCoPilot: Enabling Agentic Pattern Making with Version-Controlled State Tracking (Supplementary Material)

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1 Study Materials

1.1 Participant Information

Tables 1 and 2 report supplementary participant information for the formative study and user study, including demographics, years of pattern-making experience (Exp.), tool accessibility in the study, and compensation. Following the grouping used in the main paper, participants with less than 0.5 years of pattern-making experience were categorized as novices, and those with 0.5 to 2 years of experience were categorized as advanced novices. Experts were senior pattern makers or educators with more than 5 years of industry experience. The AI column refers to whether AI-based pattern assistants [1, 4] are used in the study, and Style3D [3] is a professional garment CAD and 3D draping system. Checkmarks indicate that participants were able to operate each tool after a simple 10-minute training session¹.

1.2 Formative Study Interview Protocol

This section summarizes the semi-structured interview protocol used in the formative study. The protocol covered the following domains:

(1) Current Workflow and Tools

- “Can you walk me through your typical workflow when developing a new pattern?”
- “Which tools, references, or examples do you usually rely on during this process?”

(2) Learning and Knowledge Acquisition

- “How did you learn pattern making, and what helped you most?”
- “Which aspects are difficult to learn or teach?”

(3) Decomposition of Garment Changes

- “When you want to change a garment design, how do you decompose it into concrete pattern edits?”
- “How do you decide what to modify first and what should follow next?”

(4) Breakdowns and Repair Strategies

- “What are the most challenging parts of pattern making for you?”
- “How do you usually recover when a modification does not work as expected?”

¹Participants in the formative study received \$20, and participants in the user study received \$30.

Table 1: Participant information for the formative study.

ID	Group	Age	Gender	Exp.	AI	Style3D
PF1	Novice	19	F	0.0	✓	
PF2	Novice	20	F	0.0	✓	
PF3	Novice	20	F	0.0	✓	
PF4	Novice	21	M	0.0	✓	
PF5	Novice	21	F	0.2	✓	
PF6	Novice	22	F	0.0	✓	
PF7	Advanced novice	19	F	0.5	✓	
PF8	Advanced novice	20	F	1.0	✓	
PF9	Advanced novice	21	M	1.5	✓	✓
PF10	Advanced novice	22	F	2.0	✓	✓
PF11	Expert	36	F	8.0	✓	✓
PF12	Expert	41	M	10.5	✓	✓
PF13	Expert	45	F	14.0	✓	

Table 2: Participant information for the user study.

ID	Group	Age	Gender	Exp.	AI	Style3D
PU1	Novice	19	F	0.0	✓	
PU2	Novice	19	F	0.0	✓	
PU3	Novice	20	M	0.0	✓	
PU4	Novice	20	F	0.0	✓	
PU5	Novice	20	F	0.2	✓	
PU6	Novice	21	F	0.0	✓	
PU7	Novice	21	F	0.0	✓	
PU8	Novice	21	M	0.3	✓	
PU9	Novice	22	F	0.0	✓	
PU10	Novice	22	F	0.0	✓	
PU11	Advanced novice	19	F	0.5	✓	
PU12	Advanced novice	19	M	0.8	✓	✓
PU13	Advanced novice	20	F	1.0	✓	✓
PU14	Advanced novice	20	F	1.2	✓	
PU15	Advanced novice	21	F	1.4	✓	✓
PU16	Advanced novice	21	M	1.6	✓	✓
PU17	Advanced novice	21	F	1.7	✓	✓
PU18	Advanced novice	22	F	1.8	✓	
PU19	Advanced novice	22	M	1.9	✓	✓
PU20	Advanced novice	22	F	2.0	✓	✓

(5) Judgment Criteria

- “How do you judge whether a modification is structurally sound?”
- “How do you determine whether the resulting garment still matches the intended appearance or style?”

Note. Representative questions are provided for illustration. Follow-up questions were used to probe specific examples, decision points, and repair strategies raised by participants.

1.3 Task Inventory

Below we list the full task pool used in the study. AI assistants are primarily used for pattern initialization, while participants were allowed to use Style3D to adjust and refine the base pattern directly.

Simple Tasks (15).

- (1) Shorten the bodice
- (2) Lengthen the bodice
- (3) Widen the waist
- (4) Widen the hem
- (5) Broaden the shoulder width
- (6) Deepen the back neckline
- (7) Increase the neckline width
- (8) Increase collar stand height
- (9) Widen the sleeve opening
- (10) Lengthen the skirt
- (11) Shorten the skirt
- (12) Increase the pants waist circumference
- (13) Widen the pant leg opening
- (14) Increase the skirt flare
- (15) Lengthen the sleeve

Medium Tasks (10).

- (1) Raise the armhole and adjust sleeve cap matching
- (2) Transfer a waist dart to the side seam
- (3) Transfer a bust dart to a princess seam
- (4) Add pleats to the front panel and reconfigure the waist dart arrangement
- (5) Add layered structure to the skirt and adjust the flare volume
- (6) Modify the collar type, seam lines, and hems
- (7) Convert a standard sleeve into a dropped sleeve
- (8) Convert a straight sleeve into a puff sleeve
- (9) Add pleats to the knees and widen the leg hems
- (10) Add a center-front placket and restructure the front panel

Complex Tasks (5).

- (1) Create a dress pattern from a reference image (fashion sketch)
- (2) Create a dress pattern from a reference image (garment photograph)
- (3) Create a coat pattern from a reference image (fashion sketch)
- (4) Create a coat pattern from a reference image (garment photograph)
- (5) Create a dress pattern from a text description

2 System Details

2.1 Interface Overview

In TailorTrace (Figure 1), experts edit garment patterns using native CAD operations while the system records the resulting semantic operations as traces in the background. They can search existing traces in the repository, commit intermediate design states, create branches from selected revisions, and reload previous versions into the CAD workspace for further editing. Each committed state is validated and stored alongside its 2D pattern geometry, 3D preview, and captured operation sequence, displayed side by side to support revision comparison and reuse. The collected expert traces then serve as training data for TailorCoPilot.

In TailorCoPilot (Figure 2), novices specify their design requirements through the dialog panel and receive an expert-guided adjustment plan. They can apply or modify symbolic parameters while inspecting the updated 2D patterns, and use the 3D draping and

pressure heatmap views to evaluate the resulting geometry and fit. This supports an iterative workflow of requirement specification, pattern adjustment, and result evaluation.

2.2 Pattern Representation

Figure 3 illustrates the structured sewing-pattern representation used in TailorCoPilot. Each pattern is represented as a set of typed entities, such as panels, edges, points, and darts, each associated with a unique identifier and linked through parent-child references. This representation provides the structural schema on which symbolic editing operations and state tracking are defined.

2.3 TailorTrace Log Example

Figure 4 presents a representative excerpt of the raw TailorTrace log, illustrating how editing actions are recorded during pattern making. The log preserves low-level operation records exported from the underlying environment.

2.4 Symbolic Operators

This appendix presents representative symbolic operators implemented in TailorTrace and used by TailorCoPilot. These operators illustrate the executable action space through which high-level reasoning outputs are compiled into concrete transformations over sewing-pattern states. It covers local geometric edits, edge and stitch operations, piece-structure modifications, topological restructuring, and history-aware control, thereby providing the operational interface between symbolic plans and editable pattern outcomes, as shown in Figure 5.

2.5 Base Pattern Retrieval

The retrieval library contains 120 editable base garments across 6 coarse categories, including tops, dresses, skirts, trousers, jackets, and coats. Each item consists of a validated sewing-pattern state, front-view rendering, garment metadata, and category label. Library patterns were internally authored and manually checked to ensure that they were editable, sewable, and compatible with the shared simulation setup.

We use RADIO [2] as the visual retriever for pattern-library initialization. All library garments are rendered under a fixed retrieval setup with a shared avatar, camera configuration, background, and lighting, so that nearest-neighbor search is driven primarily by garment silhouette and local design cues rather than rendering variation. Each library item is indexed with front view, and retrieval is performed by cosine similarity in the RADIO embedding space. At inference time, the system retrieves the top-4 candidates and uses the highest-ranked valid item as the initial editable base pattern. For text-only inputs, retrieval is first restricted to a coarse garment category predicted from the user request, after which RADIO is applied within the filtered subset. When the similarity score falls below a preset threshold τ_r , the interface returns the top-4 candidates for user selection.

2.6 Cloth Simulation Settings

To ensure consistency across experiments, all simulations use a shared set of cloth parameters, summarized in Table 3.

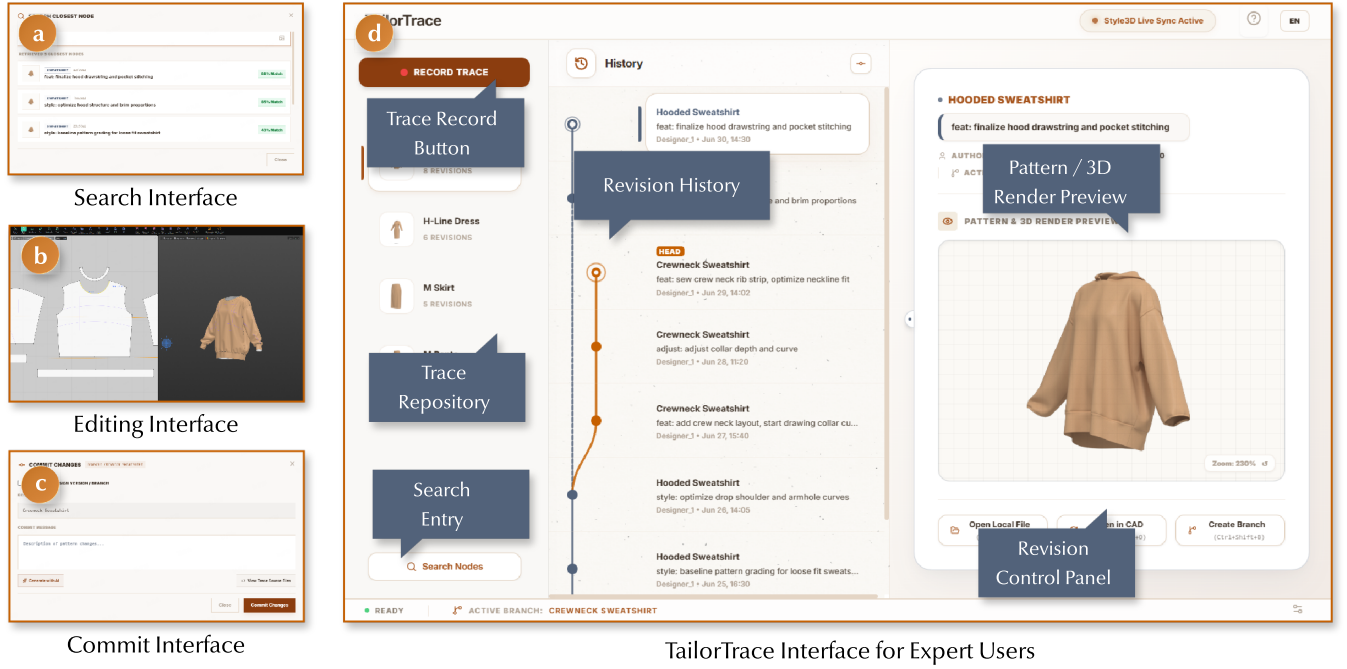


Figure 1: Overview of the TailorTrace interface for expert users. (a) Search interface for locating entries in the trace repository. (b) Editing interface for exploring garment designs using CAD software while recording operation traces. (c) Commit interface for submitting the current design state. (d) Main TailorTrace interface, comprising trace records, a search entry, revision history, 2D pattern and 3D render previews, and a revision control panel.

Table 3: Cloth simulation parameters used across experiments.

Category	Parameter	Value
Stretch	Stretch Stiffness	15
Bend	Bend Stiffness	20
General	Areal Density (GSM)	66.67
General	Thickness (mm)	0.14
Advanced	Bend Damping Ratio	0
Advanced	Bend Buckling Ratio	0
Advanced	Dynamic Friction Coefficient	0.03
Advanced	Static Friction Coefficient	0.03
Structural	Particle Distance (mm)	6.00
Structural	Layers	0
Structural	Warp Density (%)	100.00
Structural	Weft Density (%)	100.00
Structural	Additional Thickness (mm)	0.00
Structural	Collision Thickness (mm)	1.50

2.7 Prompting Architecture

We implement TailorCoPilot as a modular prompting architecture with progressive disclosure. Instead of presenting all instructions, operator definitions, and repair heuristics in a single monolithic prompt, the system maintains a compact core prompt and conditionally attaches stage-specific modules only when they are needed at inference time. This design reduces prompt bloat while preserving stage-specific precision and keeping model outputs aligned with executable symbolic operators.

Core prompt. The core prompt provides the persistent guidance shared across all stages, including the model role, the structured sewing-pattern state schema, general validity constraints, and the high-level policy for symbolic operator usage.

Planning module. When the system translates a target intent and the current pattern state into an executable operator sequence, it activates a planning prompt that requests a stepwise plan grounded in the current state, the available operator vocabulary, and optional retrieved references. This module asks the model to identify the main discrepancies between the current state and the target intent, decompose the transformation into interpretable local edits, preserve cross-piece consistency, and make stitch updates explicit when needed.

Referenced resources. To keep the shared prompt context compact, detailed operator signatures, state-field descriptions, formatting constraints, and representative examples are maintained as lightweight referenced resources and introduced only when required by the active stage.

Overall, this modular design supports a planning, validation, and repair loop while avoiding unnecessary duplication across prompts.

2.8 TailorCoPilot Dataset Generation

The TailorCoPilot dataset was constructed from validated TailorTrace state transitions after applying a trace-level train/validation split. For each sampled source-target pair, Gemini was used to draft a natural-language design-intent description, which domain

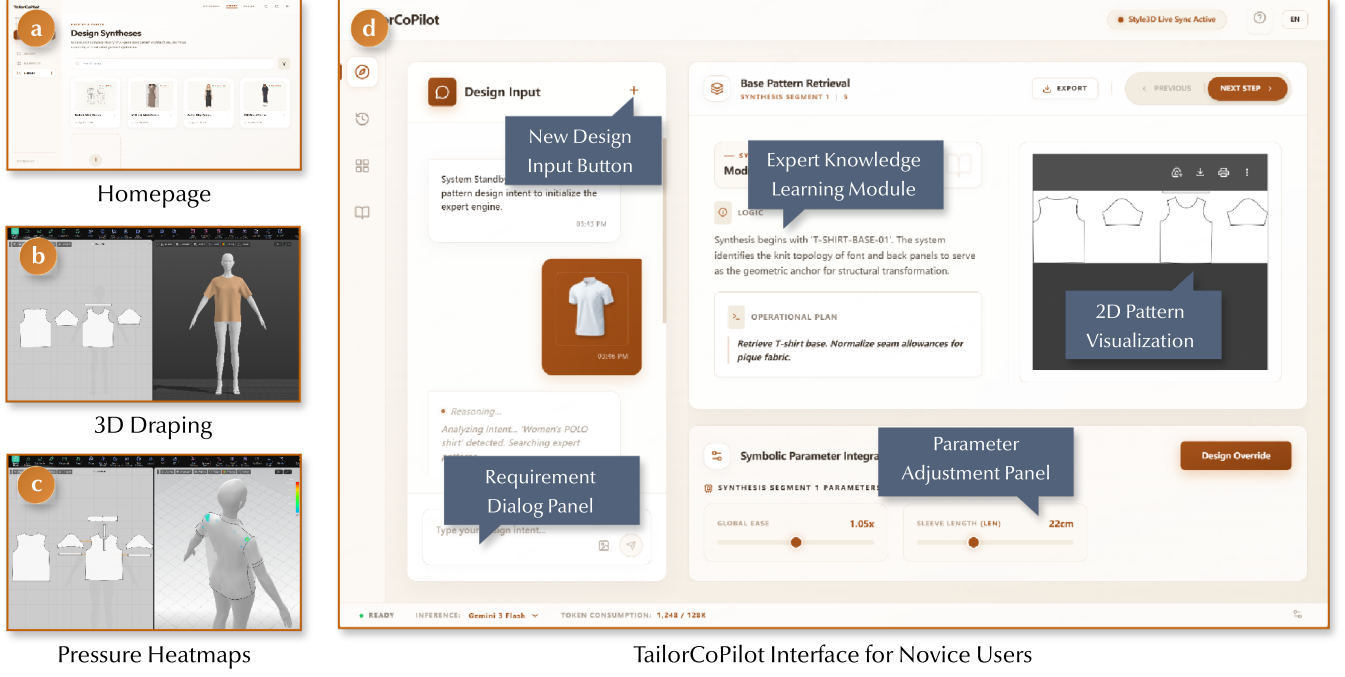


Figure 2: TailorCoPilot interface overview. (a) Homepage of TailorCoPilot. (b) 3D draping workspace for inspecting garment geometry. (c) Pressure heatmap view for assessing local fit and tension. (d) Main interface for novice users, integrating a requirement dialog panel, an expert knowledge learning module, 2D pattern visualization, and a parameter adjustment panel.

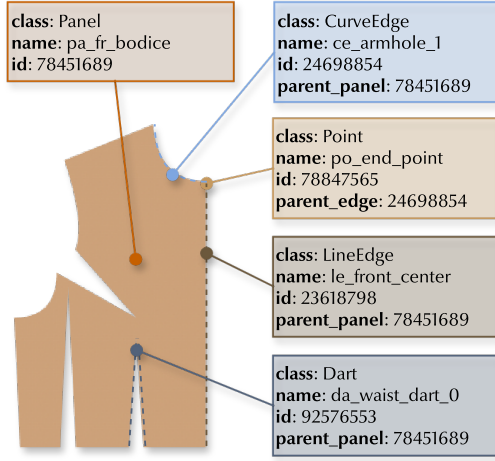


Figure 3: Structured sewing-pattern state representation. Each pattern is encoded as typed entities, including panels, edges, points, and darts, linked by unique identifiers and parent-child relations.

experts then manually verified and corrected for technical terminology and consistency with the actual edit. The final dataset contains approximately 2,500 description-operation pairs and is used to evaluate operation synthesis on held-out traces, rather than large-scale zero-shot generalization.

3 Extended Results

3.1 Additional Statistical Results

This section reports additional model statistics and interaction tests. For task success, logistic regression showed that none of the tested interactions involving Workflow reached significance (all $p > .20$).

For completion time, the main effect of Workflow was significant, $F(1, 238) = 14.72$, $p < .001$, indicating a 41.7% reduction in completion time with TailorCoPilot compared to the baseline workflow. The Workflow $\times$ Difficulty interaction was also significant, $F(2, 238) = 4.11$, $p = .018$. All other interactions involving Workflow were not significant (all $p > .30$).

For artifact quality, the main effect of Workflow was significant, $F(1, 238) = 11.94$, $p < .001$, with an estimated marginal mean difference of 1.85 (95% CI [0.93, 2.77]) on the artifact quality scale in favor of TailorCoPilot. No significant interactions were found (all $p > .24$).

For NASA-TLX, the main effect of Workflow was significant, $F(1, 238) = 30.25$, $p < .001$, indicating lower perceived workload with TailorCoPilot, corresponding to a Cohen's d of 0.71. The Workflow $\times$ Expertise interaction was not significant ($p > .30$). Figure 6 presents descriptive NASA-TLX profiles by expertise group and difficulty level using an adapted 5-point scale. Across expertise levels and difficulty conditions, TailorCoPilot generally resulted in lower mean scores across all six NASA-TLX dimensions compared with the baseline.

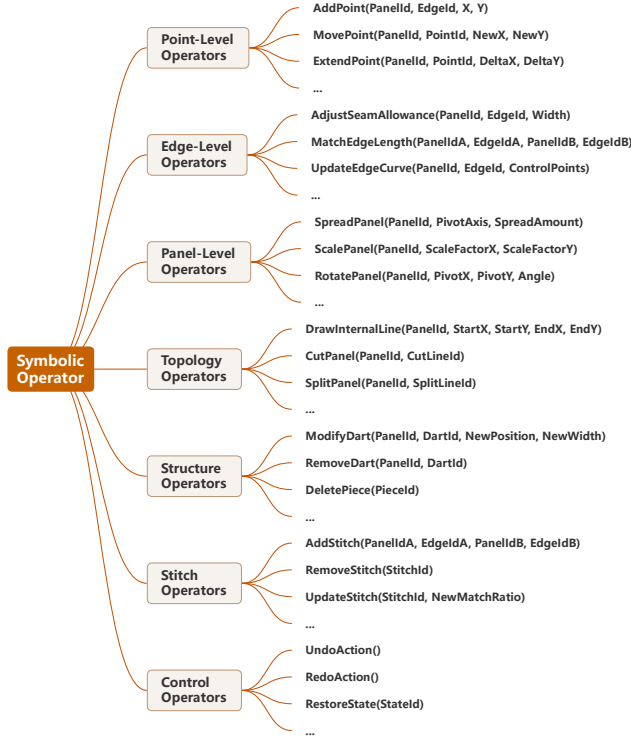


Figure 5: Representative symbolic operators used in TailorTrace and TailorCoPilot. The operator set provides the executable transformation layer that maps symbolic editing plans to concrete pattern-state revisions.

```

"Command History": [
  {
    "Description": "Transform Multi Mesh",
    "Name": "Move Pattern",
    "Parameters": "ClothPiece: 152537120 Pre-prod Sample.Body Tube.L Symm
  },
  {
    "Description": "Pattern Editing",
    "Name": "Pattern Editing",
    "Parameters": "ClothPiece: 152545404 San Jia - Front, Vertex: Vertex
  },
  {
    "Description": "Cut Pattern",
    "Name": "Cut Pattern",
    "Parameters": "Need Sew: false; Pattern: SDPattern47; Cut Edges: Edge
  },
  {
    "Description": "Pattern Scaled",
    "Name": "Move Pattern",
    "Parameters": "ClothPiece: 152537120 Pre-prod Sample.Lower Collar Sta
  },
  {
    "Description": "Pattern Scaled",
    "Name": "Move Pattern",
    "Parameters": "ClothPiece: 152537120 Pre-prod Sample.Lower Collar Sta
  },
  {
    "Description": "Transform Multi Mesh",
    "Name": "Move Pattern",
    "Parameters": "ClothPiece: 152537120 Pre-prod Sample.Lower Collar Sta
  },
  {
    "Description": "Pattern Editing",
    "Name": "Pattern Editing",
    "Parameters": "ClothPiece: 152545404 San Jia - Front_1, Vertex: Verte
  },
  {
    "Description": "Pattern Editing",
    "Name": "Pattern Editing",
    "Parameters": "ClothPiece: 152545404 San Jia - Front_1, Vertex: Verte
  },
]
  
```

Figure 4: Representative excerpt of the raw TailorTrace log format.

3.2 Additional Cases

We include additional qualitative examples selected as representative cases from successful trials to illustrate typical system behavior across different task complexities, as shown in Figure 7.

References

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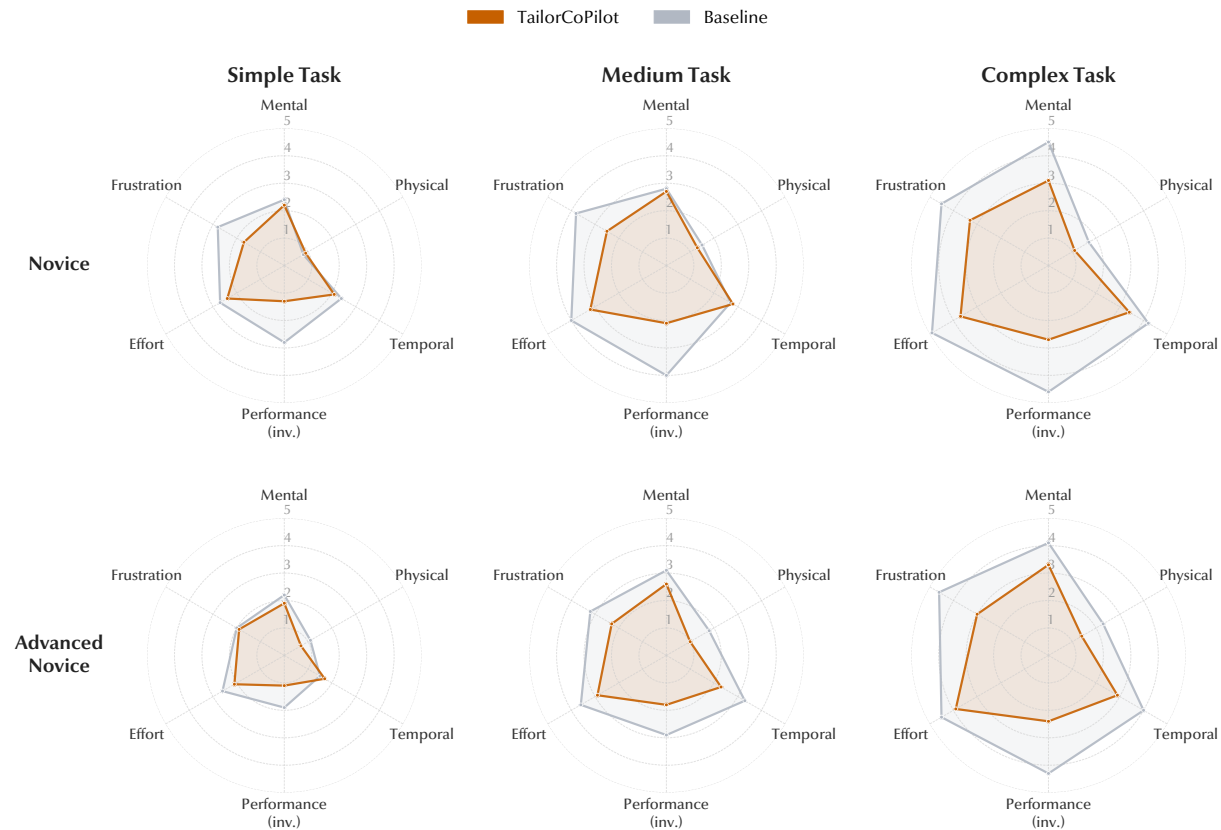


Figure 6: Descriptive NASA-TLX profiles by workflow, task difficulty, and expertise group. Radar plots show mean ratings on the six NASA-TLX dimensions for novice and advanced novice participants under TailorCoPilot and the baseline. Across both expertise groups, the profiles suggest generally lower perceived workload under TailorCoPilot.

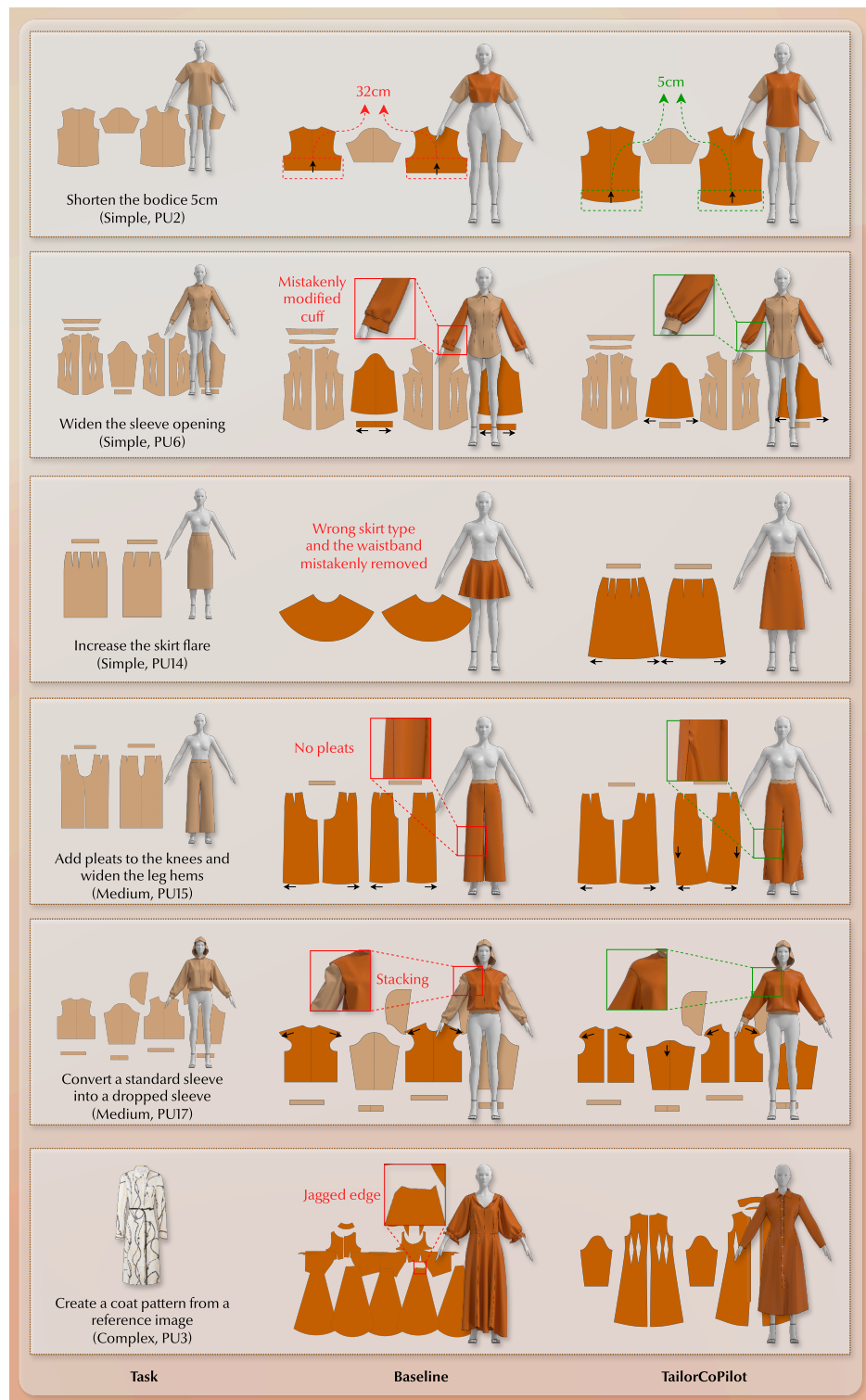


Figure 7: Qualitative comparison across simple, medium, and complex tasks. Each row presents the input (base patterns or reference images), baseline outputs with typical errors, and improved results from TailorCoPilot.